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ORIGINAL ARTICLE

Lower body blood flow restriction training may induce remote muscle strength adaptations in an active unrestricted arm

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Abstract

Purpose We examined the concurrent characteristics of the remote development of strength and cross-sectional area (CSA) of upper body skeletal muscle in response to lower body resistance training performed with an applied blood flow restriction (BFR).

Methods Males allocated to an experimental BFR group (EXP; $n = 12$) or a non-BFR control group (CON; $n = 12$) completed 7-weeks of resistance training comprising three sets of unilateral bicep curls [50% 1-repetition maximum (1-RM)], then four sets of bilateral knee extension and flexion exercises (30% 1-RM). EXP performed leg exercises with an applied BFR (60% limb occlusion pressure). 1-RM strength was measured using bilateral leg exercises and unilateral bicep curls in both trained and untrained arms. Muscle CSA was measured via peripheral quantitative computed tomography in the dominant leg and both arms.

Results 1-RM in the trained arm increased more in EXP (2.5 ± 0.4 kg; mean ± SEM) than the contralateral untrained arm (0.8 ± 0.4 kg), and the trained arm of CON (0.6 ± 0.3 kg, $P < 0.05$). The increase in knee extension 1-RM was twofold that of CON ($P < 0.01$). Knee flexion 1-RM, leg CSA, and trained arm CSA increased similarly between groups ($P > 0.05$), while untrained arm CSA did not change ($P > 0.05$).

Conclusion Lower limb BFR training increased trained arm strength more than the contralateral untrained arm, and the trained arm of controls. However, there was no additional effect on muscle CSA. These findings support evidence for a BFR training-derived remote strength transfer that may be relevant to populations with localised movement disorders.

Keywords Blood flow restriction · Strength · Hypertrophy · Resistance · Remote

Abbreviations

1-RM	One-repetition maximum
4w	Week 4 of the training programme
8w	Week 8 of the training programme
ACSM	American College of Sports Medicine
ANOVA	Analysis of variance
BFR	Blood flow restriction
BMI	Body mass index
CON	Non-blood flow restriction training group
CON-T	Trained arm of non-blood flow restriction training group
CON-U	Untrained arm of non-blood flow restriction training group
CSA	Cross-sectional area
EXP	Blood flow restriction resistance training group
EXP-T	Trained arm of blood flow restriction resistance training group
EXP-U	Untrained arm of blood flow restriction resistance training group
LOP	Limb occlusion pressure
pQCT	Peripheral quantitative computed tomography

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Introduction

Skeletal muscle hypertrophy and strength adaptations with resistance training are generally localised to the working limb (Tesch 1988; Young et al. 1983; Moss et al. 1997).

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However, less pronounced strength gains may also occur in the contralateral limb due to the phenomenon known as cross-transfer (or cross-education) (Scripture et al. 1894; Munn et al. 2004). Cross-transfer is particularly evident following unilateral resistance training. Among other hypotheses, it is suggested to be the result of reorganised central and/or peripheral motor pathways projecting to the contralateral limb to effect a more coordinated neural drive to untrained muscles (Lee and Carroll 2007). While cross-transfer between contralateral limbs is well accepted (Munn et al. 2004), transfer of adaptations to other remote sites, such as from muscles of the lower body to upper body, has not been the subject of many investigations (Madarame et al. 2008; West et al. 2010; Rønnestad et al. 2011; Walker et al. 2004; Hansen et al. 2001). This is surprising as remote strength transfer from one trained limb to another inactive/less active limb would be an important interventional outcome for populations with localised movement disorders, such as those in the early phases of musculoskeletal rehabilitation.

While neurological adaptations increase muscle strength with resistance training, skeletal muscle hypertrophy may also contribute to increased muscle strength following chronic long-term resistance training (Folland and Williams 2007). Evidence for remote transfer of muscle adaptations due to resistance training is unclear, though unlike cross-education, it may require the remote target muscle to also be active during exercise sessions. Lower body resistance exercise induced a significantly greater change in bicep curl one-repetition maximum (1-RM) strength (Rønnestad et al. 2011) and isometric strength (Hansen et al. 2001) of a trained arm compared with those that underwent only arm training (i.e., without leg resistance training). Improved arm muscle cross-sectional area (CSA) from baseline for the group that underwent leg and arm training combined, which was not seen for the group that underwent arm training alone, has also been observed (Rønnestad et al. 2011). However, the legitimacy of this result has been questioned (Phillips 2012), and others have failed to demonstrate any observation of remote transfer of muscle adaptations following resistance training (West et al. 2010; Walker et al. 2004).

Blood flow restriction (BFR) training has also been implicated in remote transfer from the lower to upper limbs (Madarame et al. 2008). Ten weeks of knee flexion and knee extension training was completed at 30% 1-RM with, or without, BFR applied proximally to both upper thighs. In addition, both groups also performed non-BFR dumbbell bicep curls in a single arm (50% 1-RM) during the same sessions, while the other arm served as a non-training control. A remote transfer effect was reported for both isometric elbow flexion torque and muscle CSA for the trained arm of the group that performed BFR leg exercise training in conjunction with the arm training (Madarame et al. 2008). It was suggested that the remote transfer effect may have been caused by greater circulating concentrations of anabolic stimuli (e.g., growth hormone, noradrenaline), associated with the recruitment of a greater muscle mass during resistance exercise of the legs when combined with BFR. However, it was acknowledged that the study was underpowered for measures of circulating anabolic stimuli. As such, when combined with variable results for the traditional resistance training, the remote transfer effect with BFR training requires further examination.

Given the potential advantage of using light loads during BFR resistance training and the possibility of a beneficial remote transfer of muscle strength and size for populations with localised movement disorders, we aimed to further elucidate whether BFR training of the lower limbs could induce remote transfer of muscle size and strength to a trained arm.

Materials and methods

Subjects

Twenty-four recreationally active young men were recruited for this study. Participants were matched for bilateral knee extension 1-RM strength and then randomly allocated to a BFR resistance training group (EXP; $n = 12$) or a non-BFR training control group (CON; $n = 12$) (Table 1). All participants had not undertaken any resistance exercise within the previous 6 months and did not present with any musculoskeletal, neurological, or vascular disease/injury. Prior to inclusion in the study, all participants provided written informed consent and underwent a pre-screening procedure including a health questionnaire. Participants were excluded if presenting with diagnosed diabetes mellitus, hypertension, or if taking medication prescribed for blood pressure control. This study was approved by the Deakin University Human Ethics Advisory Group (Project number 2013-296).

Experimental design

Participants attended the laboratory for 24 sessions over 8 weeks (Fig. 1). These comprised an initial familiarisation session followed by a testing session at least 3 days later (Baseline) to measure unilateral 1-RM strength on both arms individually, and bilateral 1-RM strength in the legs, while measurements of CSA were taken in both arms and the dominant leg. After a further 3–5 days, participants then commenced the training programme that comprised 20 resistance exercise sessions in which unilateral bicep curls were performed in the dominant arm only, followed by bilateral knee extension and bilateral knee flexion exercises in each training session. The non-dominant arm was used as an untrained control. For EXP, during all leg exercises, BFR was applied around the most proximal

Table 1 Subject characteristics, repetition completion, and limb occlusion pressure (LOP) variables

Variable	EXP	CON
Age (years)	22.6 ± 3.3	22.1 ± 2.5
Height (cm)	177.5 ± 9.5	174.1 ± 6.7
Weight (kg)	73.0 ± 13.6	72.4 ± 11.2
BMI (kg/m^2)	23.0 ± 2.6	23.8 ± 3.0
Knee extension repetition completion (%)	99 ± 1	100 ± 0
Knee flexion repetition completion (%)	100 ± 0	100 ± 0
Elbow flexion repetition completion (%)	100 ± 0	100 ± 0
Baseline LOP (mmHg)	203 ± 22	199 ± 23
Training session 1–10 restriction pressure (60% LOP; mmHg)	122 ± 13	
4w LOP (mmHg)	204 ± 18	
Training session 11–20 restriction pressure (60% LOP; mmHg)	122 ± 11	
Resting systolic blood pressure (mmHg)	117 ± 9	116 ± 8
Resting diastolic blood pressure (mmHg)	75 ± 8	77 ± 8
Limb circumference under BFR cuff (cm)	54.3 ± 4.8	56.3 ± 5.3
Vertical distance between blood pressure cuff and BFR cuff (cm)	26.8 ± 4.0	28.0 ± 3.4

Data are presented as mean ± SD

4w week 4 of the training programme, *BFR* blood flow restriction, *CON* non-BFR training control group ($n = 12$), *EXP* BFR resistance training group ($n = 12$)

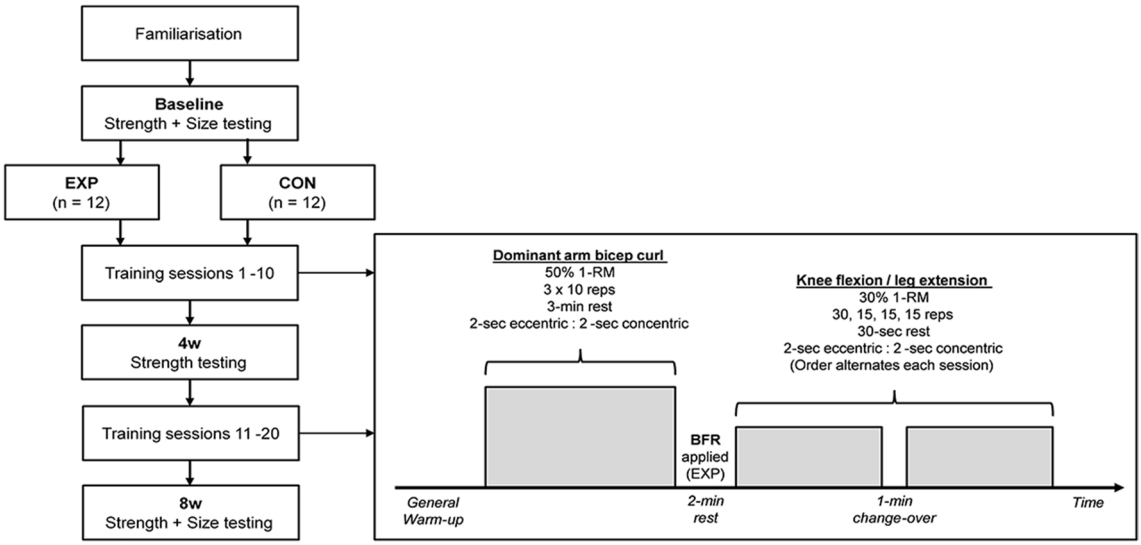

Fig. 1 Study protocol. *1-RM* one-repetition maximum, *4w* week 4 of the training programme, *8w* week 8 of the training programme, *BFR* blood flow restriction, *CON* non-BFR training control group, *EXP* BFR resistance training group

portion of both legs and inflation pressure was maintained until completion of all leg exercises for the training session. After ten training sessions, a mid-testing session was used to re-examine muscle strength only. This occurred in week 4 of the programme (4w). Three to five days following completion of all 20 training sessions, a final post-testing session was performed to measure muscle strength and size. This occurred in week 8 of the programme (8w).

Familiarisation

Familiarisation began with a pre-screening procedure. Participants were then introduced to the laboratory space and the equipment to be used throughout the training programme. After this, participants were instructed on the exercises to be used in the study and the machinery on which the lower limb exercises would be performed. This was

followed by a familiarisation with BFR exercise that comprised light bilateral dumbbell bicep curl exercises where the dominant arm had BFR applied and the non-dominant arm did not. This was performed in the arms to allow participants to experience the perceptual comparison between BFR and non-BFR exercise when the lifted load is identical. While BFR training was performed in the legs only during the following training sessions, this was not performed on the legs during familiarisation to minimise any impact on the outcomes from the future training sessions.

Training protocol

All participants were required to perform 20 sessions of resistance training with a frequency of three sessions per week. Loads for training sessions 1–10 were calculated as a percentage of 1-RM measured during Baseline strength testing, and sessions 11–20 from the 4w strength testing session. Training sessions were identical for EXP and CON groups (Fig. 1) apart from the application of BFR during all leg exercises for EXP. Sessions began with a general 5-min warmup followed by three sets of dumbbell bicep curl exercises performed in the dominant arm only. Participants performed ten-repetition sets at a load of 50% 1-RM separated by 3-min rest periods. This volume, load, and rest interval was prescribed based on the previous research performed by Madarame et al. (2008). Following this, participants performed four sets of seated bilateral knee extension and knee flexion exercises, the order of which alternated every training session. Both knee extension and knee flexion exercises were performed at 30% 1-RM and comprised a set of 30 repetitions, followed by three sets of 15 repetitions separated by 30-s rest periods. This was followed by a 5-min general cool-down. Arm exercise preceded leg exercise to avoid a possible confounding effect of central fatigue on elbow flexion performance. All lifts were performed at a 2-s eccentric and 2-s concentric cadence, guided by metronome.

Muscle strength

All 1-RM strength testing followed the American College of Sports Medicine (ACSM) guidelines for assessment of maximal muscle strength (Thompson et al. 2010). 1-RM strength testing was preceded by a general 5-min warmup and then a specific submaximal warmup for all following 1-RM exercises, and determined using the same posture and machinery employed during the training programme. Elbow flexion 1-RM strength was assessed in arms individually by unilateral bicep curls using a dumbbell. Knee extension and knee flexion 1-RM strength was assessed with bilateral lifts. Knee extension exercises were performed using a seated knee extension machine (Nautilus Nitro® Plus leg extension, Nautilus Commercial Fitness, VA, USA) and knee flexion with a seated knee flexion machine (Nautilus Nitro® Plus seated leg curl, Nautilus Commercial Fitness, VA, USA). Acceptable attempts for each exercise required the participant to lift through the full range of motion in a controlled manner. Participants were permitted to grip the handles of the exercise machines at all times for both training and testing sessions, while fixation belts were not used.

Muscle cross-sectional area

Muscle CSA was measured at Baseline and post-training in the dominant leg and both arms by use of peripheral quantitative computed tomography (pQCT) (Stratec XCT3000, Stratec Medizintechnik, Baden-Württemberg, Germany). All scans were performed by the same researcher. Lower limb scans were obtained at 25% of the dominant leg femur length (Fig. 2a) (Coupaud et al. 2009; Seynnes et al. 2007), and upper limb scans at 50% of humerus length (Fig. 2b) for both dominant and non-dominant arms (Shepstone et al. 2005). All measurements were performed with the participant laying supine with the limb to be scanned positioned through the centre of the pQCT gantry. Participants were asked to remain still and to breathe normally during the scan acquisition. Lower limb scans were acquired with a voxel size of 0.5 mm and upper limb scans with a voxel size of 0.35 mm.

pQCT images were analysed using the BoneJ plugin for ImageJ, a public domain image analysis software (Doube et al. 2010). Muscle tissue (muscle CSA, mm^2) was defined as voxels with a density > 40 and < 200 mg/cm^2 (Rantalainen et al. 2014) and arm rests were removed from images; then, total muscle CSA was calculated for all soft tissue within those thresholds. In a separate analysis, groups of knee flexor and extensor muscles were independently analysed for CSA at 25% of femur length. However, subsequent statistical analyses showed identical effects for the flexor and extensor groups to that for total muscle CSA. Therefore, only total muscle CSA is reported.

Blood flow restriction

For EXP, during all training sessions, blood flow was restricted to the legs using an automatic tourniquet system (ATS 3000, Zimmer Inc., OH, USA) connected to inflatable pneumatic cuffs (86-cm length, 10.5-cm width, and 8-cm bladder width). Following completion of bicep curl exercises, cuffs were placed around the most proximal portion of both thighs and BFR was applied continuously throughout, and between, both knee extension and knee flexion exercises. Pressure was released immediately following completion of the final set of all leg exercises. The total time under restriction for EXP participants was roughly 14 min.

Prior to warmup on the first training session, individualised applied BFR pressures were calculated as 60% of

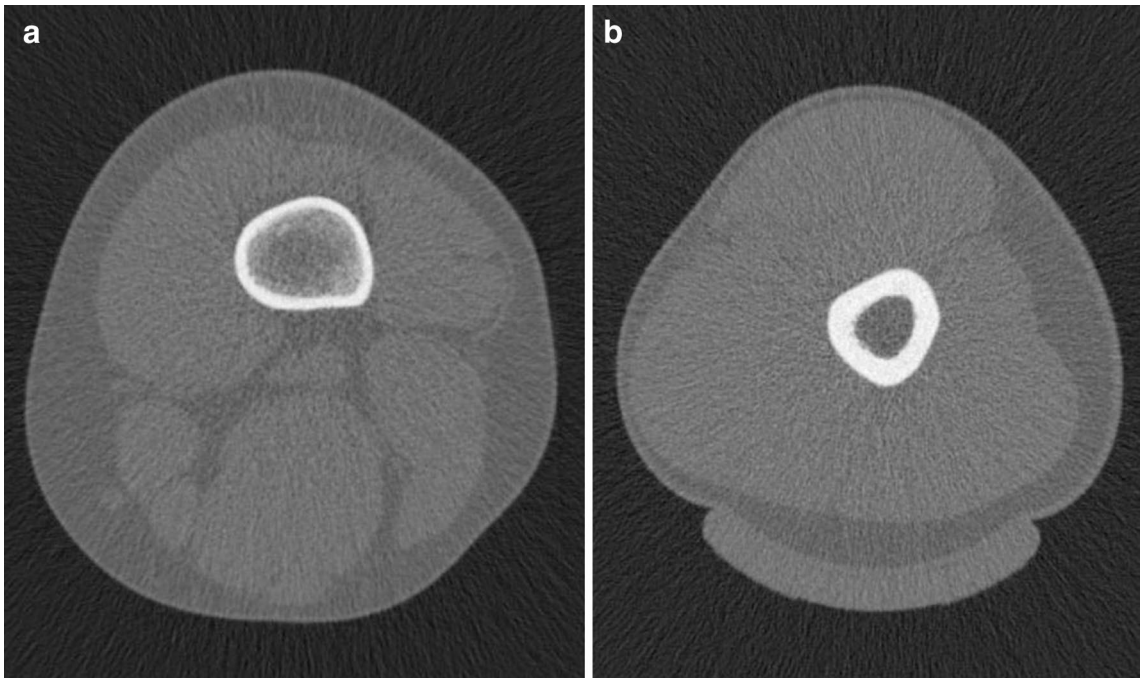

Fig. 2 Representative images of cross sections at 25% length of the femur (**a**) and 50% length of the humerus (**b**)

the participant's limb occlusion pressure (LOP; Table 1), a method employed previously during leg resistance exercise (Staunton et al. 2015). This pressure was then applied to EXP participants for the following ten training sessions. This process was repeated for EXP participants at the beginning of the 11th training session to be used for all remaining training sessions. This was conducted with participants resting in a seated position with BFR cuffs applied to both upper thighs. This posture for LOP measures was similar to that used for the following leg training. Participants sat on the edge of the chair to avoid resting on the cuffs as this could possibly influence measured LOP. With restriction cuffs in place on the limb, a plethysmograph (LOP Sensor Kit, Zimmer Inc., OH, USA) was applied to the second toe of the foot. Following 5-min rest, the LOP measurement was performed by the automatic tourniquet system using an inbuilt function. Pressure in the restriction cuffs increased gradually until tissue blood flow was no longer detected in the toe. Cuffs would then deflate and LOP was recorded. LOP measurements were conducted twice on each limb, and if all were within 20 mmHg, the average was used to estimate 60% LOP to be applied during leg training exercises. LOP measurements greater than 20 mmHg apart required another test to be conducted, after which the average of all tests was used to set cuff pressure for leg training.

Data presentation and statistical analysis

Unless otherwise stated, all data are presented as mean ± SEM. Anthropometry, repetition completion rates, and restriction pressure variables were compared between training groups (CON; EXP) with unpaired *t* tests assuming equal variance. Baseline and 4w LOP measurements were compared with paired *t* tests.

Absolute leg strength data (1-RM) were analysed using a mixed-model analysis of variance (ANOVA) for GROUP (CON; EXP) and TIME (Baseline, 4w and 8w). The change in absolute leg 1-RM strength from Baseline was instead examined using an analysis of covariance (ANCOVA) to adjust for absolute 1-RM strength values at Baseline. For both these analyses, any significant main effects or interactions were further examined for specific differences using a Fisher's projected least significant difference *post hoc* test. Arm strength data (1-RM) were similarly analysed (GROUP × TIME × ARM – Trained; Untrained).

Absolute leg muscle CSA data were analysed using a mixed-model ANOVA for GROUP and TIME (Baseline, 8w). The change in absolute leg muscle CSA from Baseline was analysed using an ANCOVA to adjust for absolute muscle CSA at Baseline. Arm muscle CSA data were similarly analysed (GROUP × TIME × ARM).

For all analyses, the significance level was set at $P < 0.05$. Statistical analyses were conducted using IBM SPSS Statistics V24.0 (IBM Corp., NY, USA).

Results

Participant characteristics are shown in Table 1. Anthropometric variables, repetition completion rates, LOP, resting blood pressure, limb circumference, and cuff distance measures were not different between groups ($P > 0.05$). LOP for EXP was also similar at Baseline and 4w ($P = 0.90$).

Muscle strength

At Baseline, both knee extension and knee flexion 1-RM strength were not different between EXP and CON (Fig. 3a, c). However, elbow flexion 1-RM strength at Baseline was greater in the trained arm for CON (CON-T) compared with the untrained arm for CON (CON-U) ($P < 0.001$), while

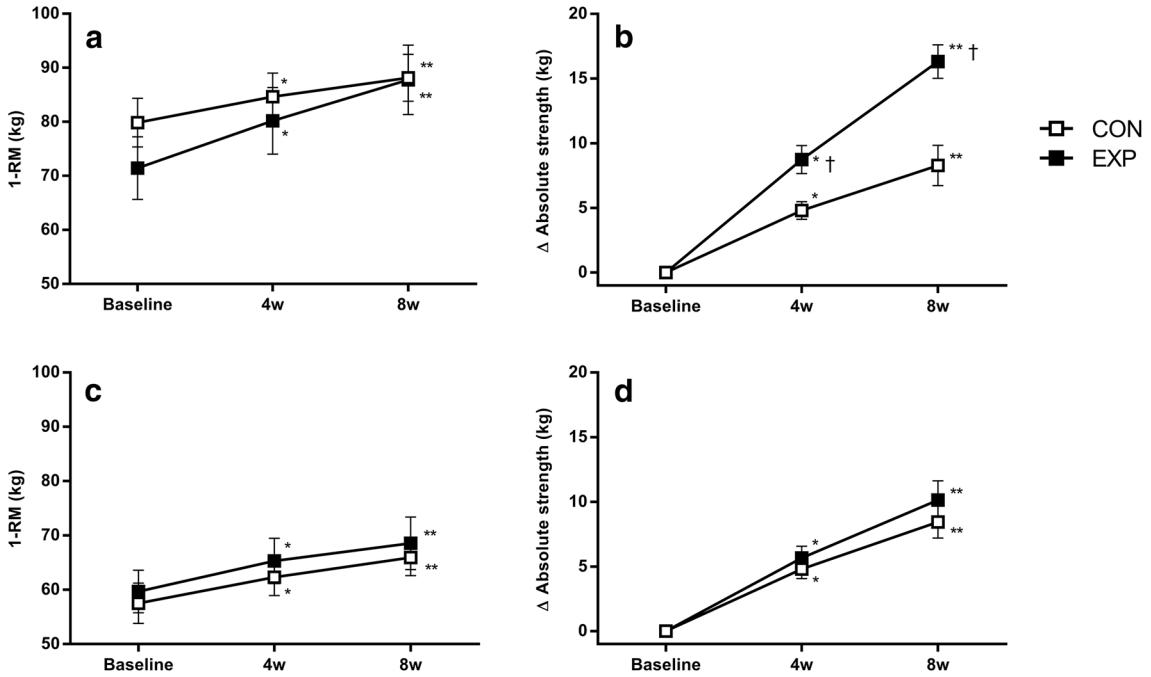

Fig. 3 1-RM strength and change in absolute 1-RM strength for bilateral knee extension (**a, b**) and knee flexion exercises (**c, d**) in blood flow restriction (EXP) and non-blood flow restriction control (CON) training groups. Data are mean ± SEM. *Different from Baseline ($P < 0.05$). **Different from 4w ($P < 0.05$). †Different from CON ($P < 0.01$)

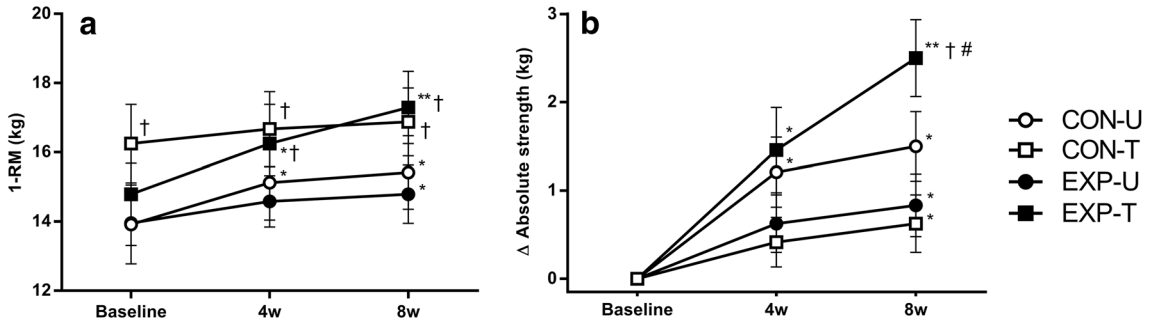

Fig. 4 1-RM strength (**a**), and change in absolute 1-RM strength (**b**) for unilateral elbow flexion exercises in trained (T) and untrained (U) arms in blood flow restriction (EXP) and non-blood flow restriction control (CON) training groups. Data are mean ± SEM. *Different from Baseline ($P < 0.05$). **Different from 4w ($P < 0.01$). †Different from Untrained arm within training group ($P < 0.05$). #Different from CON-T ($P < 0.05$)

1-RM strength at Baseline was similar between arms for EXP (Fig. 4a).

Knee extension 1-RM strength (Fig. 3a) increased significantly from Baseline to 4w in both EXP and CON ($P < 0.05$) and then increased again from 4w to 8w ($P < 0.05$). However, knee extension 1-RM strength was not different between groups at any time-point ($P > 0.05$). The change in absolute knee extension 1-RM strength from Baseline (Fig. 3b) was also progressive in EXP from 4w (8.7 ± 1.1 kg) to 8w (16.3 ± 1.3 kg), but to levels significantly greater than for CON (4w = 4.8 ± 0.7 kg; 8w = 8.3 ± 1.6 kg) ($P < 0.01$).

Knee flexion 1-RM strength (Fig. 3c) increased significantly from Baseline to 4w in both EXP and CON ($P < 0.05$) and then increased again from 4w to 8w ($P < 0.05$). However, knee flexion 1-RM strength was not different between groups at any time-point ($P > 0.05$). The change in absolute knee flexion 1-RM strength from Baseline (Fig. 3d) was also progressive in EXP from 4w (5.7 ± 0.9 kg) to 8w (10.1 ± 1.5 kg) and CON (4w = 4.8 ± 0.7 kg; 8w = 8.5 ± 1.3 kg), and was also not significantly different between groups at any time-point ($P > 0.05$).

Elbow flexion 1-RM strength (Fig. 4a) increased significantly from Baseline to 8w in the trained arm of EXP (EXP-T; Baseline = 14.8 ± 0.9 kg; 8w = 17.3 ± 1.0 kg), the untrained arm of EXP (EXP-U; Baseline = 14.0 ± 0.6 kg; 8w = 14.8 ± 0.8 kg), and CON-U (Baseline = 13.9 ± 1.1 kg; 8w = 15.4 ± 1.1 kg) ($P < 0.05$). 1-RM strength for EXP-T was significantly greater than EXP-U at both 4w and 8w ($P < 0.05$), as was 1-RM strength for CON-T compared with CON-U ($P < 0.05$). When expressed as the change in absolute elbow flexion 1-RM strength from Baseline (Fig. 4b), there was an increased strength for all arms at 8w ($P < 0.05$), while being significantly greater for EXP-T at 8w (2.5 ± 0.4 kg) compared with EXP-U (0.8 ± 0.4 kg) ($P < 0.05$) and CON-T (0.6 ± 0.3 kg) ($P < 0.05$).

Muscle size

Muscle CSA was not significantly different between groups at Baseline for either upper or lower limbs ($P > 0.05$) (Fig. 5). Muscle CSA increased in all trained limbs for both EXP and CON. For the leg, muscle CSA was larger at 8w compared with Baseline for both EXP and CON ($P < 0.001$) (Fig. 5a). However, leg muscle CSA was not different between EXP and CON at any time-point, including when expressed as the change in absolute muscle CSA from Baseline (Fig. 5b) ($P > 0.05$).

For the arms, muscle CSA was not significantly different between EXP and CON ($P > 0.05$) (Fig. 5c). However, muscle CSA of the trained arms was larger than untrained arms ($P < 0.001$) given a larger increase in muscle CSA from Baseline to 8w in the trained arms ($P < 0.01$). This was not observed when expressed as a change in absolute muscle CSA from Baseline ($P > 0.05$) (Fig. 5d).

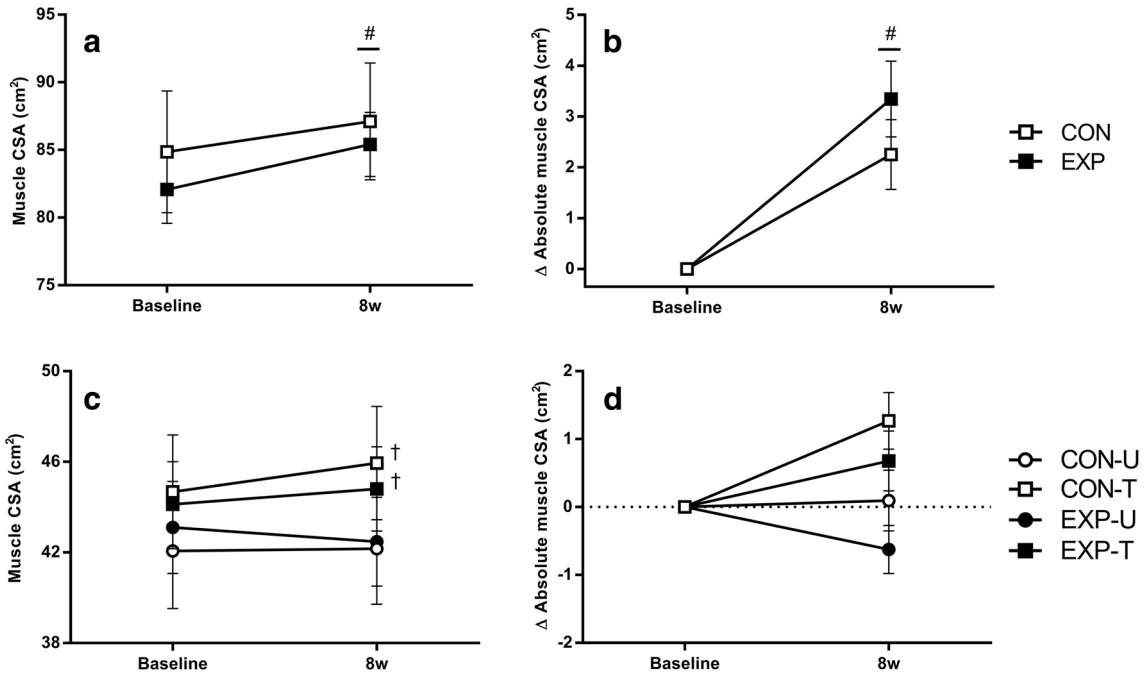

Fig. 5 Total muscle cross-sectional area (CSA), and change in absolute muscle CSA at 25% femur length (**a, b**) and 50% humerus length (**c, d**) in blood flow restriction (EXP) and non-blood flow restriction control (CON) training groups. Data are mean ± SEM. #Main effect for TIME ($P < 0.05$). †Main effect for ARM ($P < 0.001$)

Discussion

The main finding of the present study was a significantly greater absolute change in elbow flexion 1-RM strength for EXP-T compared with both EXP-U and CON-T. In addition, BFR training in the legs improved knee extension 1-RM strength of EXP twice as much as CON. This suggests that the effects of BFR training in the legs had an additional remote transfer effect on the strength adaptations in an arm that was active during the same training sessions. However, muscle CSA increased similarly between EXP and CON in the lower limb, and between EXP-T and CON-T. This suggests that the remote transfer effect observed for muscle strength was likely the result of a neuromuscular adaptation, rather than via a contribution from skeletal muscle hypertrophy.

Like the present study, a remote transfer effect for elbow flexor strength following BFR training has previously been observed by Madarame et al. (2008). However, this was only for isometric strength. Elbow flexion 1-RM strength increased similarly between trained arms for BFR and non-BFR leg training groups. The difference in all 1-RM strength outcomes between our data and that reported by Madarame et al. (2008) may be due to the volume of exercise within each training session. We prescribed one more set of 15 repetitions than for both leg exercises in what is a more common prescription (Brandner et al. 2015a; Yasuda et al. 2011; May et al. 2017). This greater training volume also increased the total time under restriction by roughly 3 min, which would equate to a 27% increase in restriction duration, assuming a similar repetition velocity and exercise changeover duration between the study performed by Madarame et al. (2008) and the present study. We also applied an individualised restriction pressure to EXP to induce a similar relative effect on blood flow for all participants, unlike Madarame et al. (2008) where an arbitrary pressure was applied (initially 160 mmHg, and increased by 20 mmHg every 2 weeks; 4 cm cuff width).

The absolute change in muscle CSA in EXP-T was 0.68 ± 0.44 cm^2 (1.64%) (Fig. 5d). This was similar to the change for CON-T (1.27 ± 0.42; 2.99%). This suggests that the significantly greater improvement in elbow flexion 1-RM strength for EXP-T compared with CON-T was most likely attributable to neuromuscular adaptations. This is surprising as the identical arm training protocol between EXP-T and CON-T would be expected to induce a similar local activation stimulus for all arm muscle groups (agonists, synergists, stabilisers, and antagonists). However, Brandner et al. (2015b) previously reported that elbow flexion exercises combined with BFR applied to the upper arm to restrict blood flow to all distal tissues (including the biceps brachii) increased the amplitude of motor evoked potentials at the corticospinal pathway for biceps brachii activation. This suggested a possible modulation of corticospinal neuroplasticity associated with the biceps brachii. Furthermore, the previous studies using non-invasive brain stimulation protocols to increase corticospinal excitability of specific hand muscles have demonstrated a spillover effect, whereby excitability of non-targeted adjacent hand muscles is also increased (Kamke et al. 2016). Therefore, it is possible that the cortical excitability at the primary control site for legs as a result of lower body BFR training in the present study may have spilled over to the proximal control site for arms, influencing motor unit recruitment of agonists in the trained arm. This strength adaptation may also be partially driven by increased circulating concentrations of anabolic factors as has been noted previously for BFR exercise (Fujita et al. 2007; Takano et al. 2005; Madarame et al. 2008; Takarada et al. 2000). However, without a measured concomitant increase in muscle CSA of EXP-T, it is unlikely that circulating anabolic factors significantly impacted on arm muscle strength/CSA in the present study.

It should be noted that cross-transfer may also have occurred between contralateral limbs. Both EXP-U and CON-U demonstrated increased elbow flexion 1-RM strength despite remaining untrained throughout the duration of the study. However, it is also possible that this improved strength in untrained arms occurred due to familiarity with 1-RM efforts as a result of repeated strength testing sessions (Schenk and Forward 1965; Ploutz-Snyder and Giamis 2001). There was also no significant difference in 1-RM strength between EXP-U and CON-U ($P > 0.05$) despite a significantly greater increase for EXP-T compared with CON-T ($P < 0.05$). This indicates that remote strength transfer may not be possible without the addition of a local mechanical tension, which supports the previous suggestions by Madarame et al. (2008).

The improvements in knee extension 1-RM strength following EXP are similar to those reported previously for BFR training (Fahs et al. 2015; Madarame et al. 2008; Karabulut et al. 2010). Knee extension 1-RM strength increased in EXP (16.3 ± 1.3 kg) by approximately twofold in comparison with CON (8.3 ± 1.6 kg). This establishes that the prescribed BFR exercise protocol had a direct local influence on the dynamic muscle strength of the restricted limb. However, there was no significant difference between EXP and CON for lower limb muscle CSA ($P = 0.66$). Lower limb muscle CSA increased by 3.34 ± 0.75 cm^2 in EXP, a result slightly lower than those reported previously for BFR leg training programmes (Madarame et al. 2008; Kubo et al. 2006). This indicates that the improvement in knee extension 1-RM strength for EXP over that of CON should be attributed to enhanced neuromuscular adaptation and not skeletal muscle hypertrophy.

The initial strength improvements in any traditional resistance training programme occur almost solely due to neuromuscular adaptations (Sale 1988), whereas BFR exercise has been shown to induce muscle hypertrophy within 3 weeks of training (Abe et al. 2006). Despite this, the difference observed in knee extension 1-RM strength between training groups in the present study was likely not derived from differences in lower limb muscle CSA. Therefore, this improved knee extension 1-RM strength may be primarily driven by a chronic increase in motor unit recruitment as it is positively correlated with strength adaptations (Häkkinen and Komi 1986; Häkkinen et al. 1985). In addition, large increases in muscle activation during BFR exercise indicate a more substantial fast-twitch fibre contribution to lifts despite the low loads used during BFR resistance exercise (Moritani et al. 1992). This is evidence for enhanced knee extensor motor unit activation during 1-RM tests in EXP compared with CON, and, perhaps, also an increased ability to coordinate synergist and stabiliser muscles (Rutherford and Jones 1986; Folland and Williams 2007), or reduced activity of antagonists (Carolan and Cafarelli 1992).

Measurement of the change in knee flexion 1-RM strength following a BFR training programme is less common than for leg extension 1-RM strength, while results are also variable (Madarame et al. 2008; Kim et al. 2009; Abe et al. 2005). Therefore, it is difficult to determine the expected effect of BFR training on dynamic strength of the knee flexors. The similar 1-RM knee flexion strength between groups in the present study but significantly greater change in knee extension 1-RM strength for EXP compared with CON may be partially attributed to the similar seated position for both exercises. It is possible that sitting on the pressurised cuffs during leg flexion exercises diffused the applied pressure applied to the active knee flexor muscles, thereby reducing the restrictive effect on the targeted muscles. This issue may not occur for the knee extensors during seated knee extension training. We suggest that future research report the posture employed for BFR exercises, and specifically knee flexion exercises (often seated or prone), to elucidate if posture may affect any expected muscle adaptation.

Limitations

Given that the response of muscle CSA (hypertrophy) can be variable within a muscle group (both between individual muscles and along the muscle length) (Narici et al. 1996, 1989; Housh et al. 1992), a limitation of the present study was that leg muscle CSA was only measured at one site (25% femur length). The intention for this selection was to more easily identify the flexor and extensor muscle groups. However, given the use of pQCT as opposed to higher resolution through MRI, this was not possible at 50% femur length. Therefore, given the lack of any difference between the flexors and extensors compared with the total leg CSA at 25%, some benefit would have been gained through an additional measurement of total muscle CSA at 50% femur length due to the larger CSA at this point and the potential for a more global snapshot of the total muscle CSA response. Therefore, we suggest that further work attempts to use the greater resolution available through MRI, and to examine multiple muscle sites to more globally capture the leg muscle CSA response.

The perceived exertion of BFR resistance exercise is often greater than non-BFR exercise of equal load (Staunton et al. 2015; Suga et al. 2009; Loenneke et al. 2013). While both CON and EXP showed equal capacity to complete the required repetitions throughout the training programme (Table 1), it was likely that the recruitment of stabilisers in conjunction with increased effort from arm muscles through gripping the handles of the leg extension/flexion machines was greater for EXP. While this could contribute to an improvement in leg strength observed in EXP, such as through greater coordination during the assessment of 1-RM, it is difficult to reconcile this effect explaining the almost double increase in strength with BFR compared with CON. In addition, coordination of stabiliser muscles is suspected to occur during the earliest phase of training for muscle strength (Rutherford and Jones 1986). While this might also have effected arm flexor strength, given that this was not evident for the untrained arms (i.e., EXP-U compared with CON-U), we expect the overall impact of any greater recruitment of stabilisers and arm muscles during the leg flexion/extension training to be negligible.

Finally, the current study and that reported by Madarame et al. (2008) only investigated remote skeletal muscle adaptations to BFR training in male healthy young adults (~22 years old). As such, while remote muscle adaptations are partially supported by the current work, it would be prudent to not generalise results to populations not included in the sample groups of either study, particularly females or clinical populations with localised movement disorders.

Conclusion

Lower limb BFR training was shown to induce a remote transfer effect on muscle strength in an active arm. This did not occur concurrently with remote muscle hypertrophy or a greater hypertrophy in EXP at the lower limb when compared with CON. Therefore, resistance training combined with BFR appears to induce remote strength transfer, primarily as a result of neuromuscular adaptations. Dynamic strength and muscle hypertrophy results were dissimilar to those reported previously (Madarame et al. 2008) but support the role of BFR for improving muscle strength of active

remote muscles. As such, further investigation of remote transfer remains warranted, especially considering the potential benefits for lower physical functioning populations such as those with localised movement disorders. A particular focus on neuromuscular adaptations should be adopted for such investigations.

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Compliance with ethical standards

Conflict of interest The authors declare no conflict of interest.

Research involving human participants All procedures performed in this study involving human participants were in accordance with the ethical standards of the institutional and/or national research committee and with the 1964 Helsinki declaration and its later amendments or comparable ethical standards.

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